

WI-FI SYNCHRONISED ANALOGUE CLOCK

WiFi-Clock

User manual for CH-899 / CHOUCHIN compatible clock movements

This movement uses your 2.4 GHz Wi-Fi network and Network Time Protocol (NTP) to set itself accurately. It normally checks time once per day, then the Wi-Fi electronics sleep between updates.

What you need

Two AA batteries, a 2.4 GHz Wi-Fi network, and a phone or computer for first-time setup.

Normal operation

After setup, the clock joins your saved network, obtains time and then returns to low-power operation.

Installation reference



Not to scale. Dimensions and component positions may vary slightly between movement revisions; check the supplied movement before cutting or drilling.

Install the movement and hands

1**Keep the movement at 12:00**

A new movement is supplied with a locking pin fitted. Leave the pin in place while mounting the movement and fitting the hands. If refitting hands later, power the movement and press **REC** to return it to its 12 o'clock reference position; wait until all movement has stopped, then remove both batteries.

2**Fit the hands**

With the movement unpowered and held at its 12 o'clock reference, fit the hour, minute and second hands so that each points exactly to 12:00. Make sure the hands are parallel and cannot touch each other, the dial or the glass.

3**Remove the locking pin**

Remove the locking pin before inserting the batteries or operating the clock. Retain the pin for future transport or hand refitting, if practical.

Important

Never insert the batteries or operate the movement while the locking pin is fitted.

Set up the clock

The setup access point is available only when the replacement firmware has no saved Wi-Fi settings - for example on first use or after Factory reset.

- 1 Wake the movement**
 After fitting the hands and removing the locking pin, insert both batteries. If no settings are saved, wait for the setup Wi-Fi network to appear. On later visits, press **M.SET** to wake the Wi-Fi module; with saved settings it joins your usual network instead of broadcasting the setup network.
- 2 Join the setup network**
 On your phone or computer, connect to **wifi-clock-setup-<chip-id>**. The final six hexadecimal characters are unique to the movement. If a captive page does not open, browse to **http://192.168.4.1/**.
- 3 Enter your settings**
 Choose or enter a 2.4 GHz Wi-Fi network, enter its password, choose an NTP server, timezone and daylight-saving setting, then choose a supported daily update time.
- 4 Save and update**
 Select **Save & update clock**. The Wi-Fi module saves the settings and restarts, connects to the selected network, obtains fresh NTP time and sends the correct local time to the clock. The setup page may become unavailable when the update completes.

Daily update and local access

Supported update times	09:00, 10:00, 21:00 or 22:00 local time. These are the values confirmed to be accepted by the CH-899 movement controller.
Station-only policy	Once Wi-Fi settings are saved, normal daily wakes do not broadcast the setup network. The movement gives saved Wi-Fi ten seconds to connect. If it cannot connect, Wi-Fi stops for that wake.
LAN address	While connected and awake, open http://wifi-clock-<chip-id>.local/ , or use the LAN IP address shown on the setup page. mDNS support varies by device and network.
NTP and fallback	A custom NTP server is tried first. If it does not respond, the firmware falls back to pool.ntp.org . If no fresh NTP response is received, the firmware does not send a new time record to the movement.

Timezone and daylight saving

Choose the closest named timezone and leave **Daylight saving** set to **Automatic** for normal use. The supplied rules include the seasonal clock changes for the selected location. Use **Disabled** only where standard time is required all year, or **Custom rule** for a valid POSIX timezone rule.

NTP supplies UTC time. The selected timezone and daylight-saving rule are applied locally before the correct local time is sent to the movement.

No external website is needed for setup

The setup page is embedded in the movement. Network access is needed only to contact an NTP server; internet access is required when that server is online rather than local.

Recovery, care and support

If a router name or password changes, a configured movement intentionally does not broadcast a fallback setup network. Use one of the recovery methods below.

Factory reset from the page When the clock is connected and awake, open its LAN page and select Factory reset . This clears the Wi-Fi and time settings, then restarts and broadcasts the setup network.	Physical reset Hold M.SET + REC together for about three seconds, then release. The movement requests a settings clear; after restart, the setup network returns. This does not reinstall vendor firmware or erase the MM32 controller.
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Quick troubleshooting

Setup Wi-Fi not visible	With saved settings, the setup network is deliberately not broadcast. To clear them, hold M.SET + REC together for about three seconds, release the buttons, then begin a fresh wake session.
Clock did not update	Check that the router still provides the saved 2.4 GHz network and that the selected NTP server is reachable. The movement will try again at its next scheduled update.
Cannot open .local address	Use the IP address displayed by the setup page. Some phones, routers and networks do not provide mDNS name resolution.
Hands are incorrect	Press REC and wait for the movement to stop at its 12 o'clock reference. Remove both batteries, refit all hands pointing exactly to 12:00, then insert the batteries again. REC alone does not clear Wi-Fi settings.

Safety, servicing and disposal

Use and care

Batteries: Use two AA batteries with matching chemistry and age; observe polarity. Do not mix rechargeable and non-rechargeable batteries. **Location:** Indoor, dry use only; keep away from heat, moisture and direct water.

Radio and servicing: Do not modify the radio module, antenna area, supply wiring or enclosure. Over-the-air updates are not provided; firmware servicing requires the documented wired procedure.

Compliance and disposal: Market-specific conformity, responsible-person and recycling information must be supplied by the retailer or importer. This manual is not a declaration of conformity. Recycle the movement and removed batteries through the appropriate local routes.

Manufacturer specifications

Model	CH-899WIFIW	Quartz accuracy	±30 seconds/month at 3.0 V and 23°C ±3°C
Drawing reference	WIFI W-01	Temperature	Operating 0 to +55°C; storage -10 to +60°C
Movement type	Wi-Fi sweep-second	Reception time	30 seconds or less
Radio band	2412-2484 MHz	Automatic reception	Once per day
Antenna	PCB antenna	Maximum correction	4 minutes 30 seconds
Batteries	2 x LR6/AA, 1.5 V	Second-hand torque	280 µN·m (about 2.86 gf·cm)
Operating voltage	2.65-3.50 V	Minute-hand torque	750 µN·m (about 7.65 gf·cm)
Claimed battery life	10 months	Operating noise	Less than 30 dB

Values reproduced from the manufacturer's CH-899WIFIW drawing. They are not independently measured and may vary between movement revisions.

Open-source firmware: Source, flashing instructions and technical documentation: github.com/maddenste/Chouchin-CH899-Firmware.

Licensed under GPL-3.0-or-later. The movement manufacturer is not affiliated with this project.

Document revision 1.0.1 - prepared September 2026. Retain this manual with the movement for future setup and recovery.